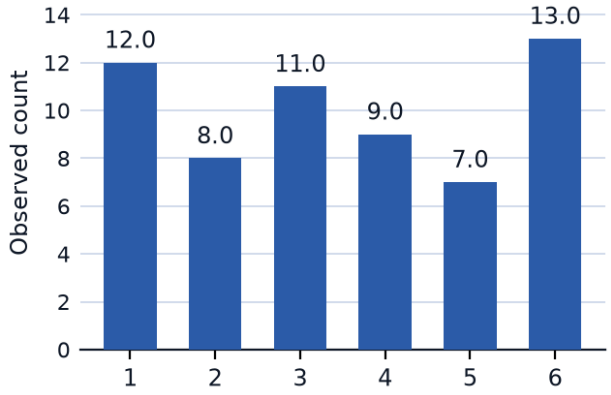
Chi-Square Tests & Regression Inference

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

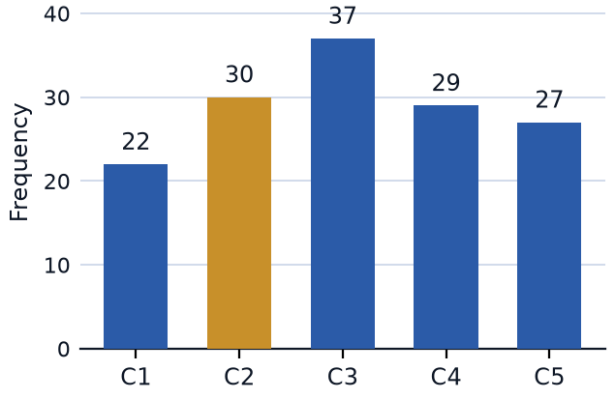
1 [Foundational] • Target: Chi-square goodness-of-fit test

A die is rolled 60 times, expecting each face to appear equally often (expected count 10 per face). The observed counts are $12, 8, 11, 9, 7, 13$. Compute the chi-square test statistic using $\chi^2 = \sum \frac{(O-E)^2}{E}$.



2 [Foundational] • Target: Chi-square goodness-of-fit test

State the null and alternative hypotheses for the goodness-of-fit test in the die-rolling scenario above.



3 [Foundational] • Target: Chi-square test for independence

A two-way table examines whether preferred learning style (visual, auditory, kinesthetic) is associated with academic major (STEM, non-STEM). State the null and alternative hypotheses for a chi-square test of independence in this context.

4 [Foundational] • Target: Chi-square test for independence

In a chi-square test of independence with a 3×2 table (3 learning styles, 2 major categories), find the degrees of freedom, using $df = (\text{rows} - 1)(\text{columns} - 1)$.

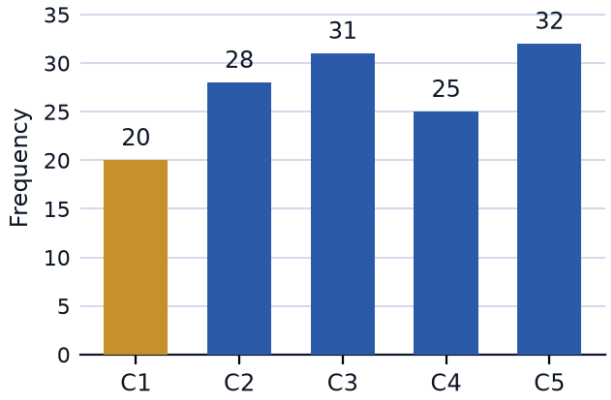
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5 [Foundational] • Target: Interpreting chi-square results

A chi-square test of independence produces a test statistic $\chi^2 = 8.2$ with $df = 2$, and a p-value of 0.017. State a conclusion at the $\alpha = 0.05$ significance level, in context.

6 [Foundational] • Target: Interpreting chi-square results

Explain a condition that must be checked before conducting a chi-square test, related to expected cell counts.



7 [Foundational] • Target: Inference for the slope of a regression line

A regression of exam score on hours studied gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.1$, based on $n = 25$ data points. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.069$ (with $df = 23$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inference for the slope of a regression line

State the null and alternative hypotheses for testing whether hours studied has a significant linear relationship with exam score, and compute the test statistic using the values from the previous problem.

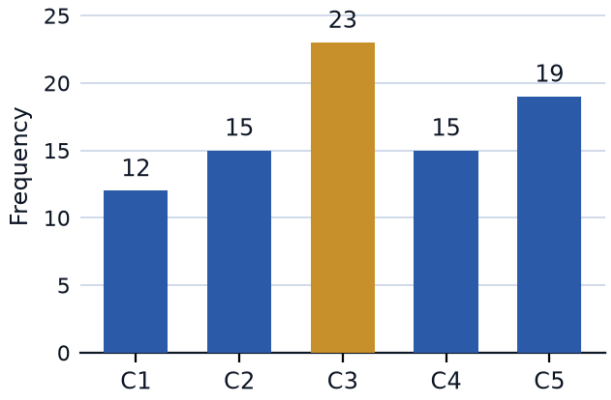
9 [Intermediate] • Target: Conditions for regression inference

State the four conditions (sometimes remembered as LINER: Linear, Independent, Normal, Equal variance, Random) required for valid inference about a regression slope.

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10 [Intermediate] • Target: Putting it together: a full inference scenario

A study finds a chi-square test statistic of $\chi^2 = 3.1$ with $df = 1$, and a critical value of $\chi^2_{0.05} = 3.84$ at the $\alpha = 0.05$ level. State the conclusion, and explain how this differs from a p-value based approach.



11 [Intermediate] • Target: Chi-square test for homogeneity

Explain the difference between a chi-square test of independence and a chi-square test of homogeneity, even though both use the same formula and table structure.

12 [Intermediate] • Target: Computing expected counts for a chi-square test

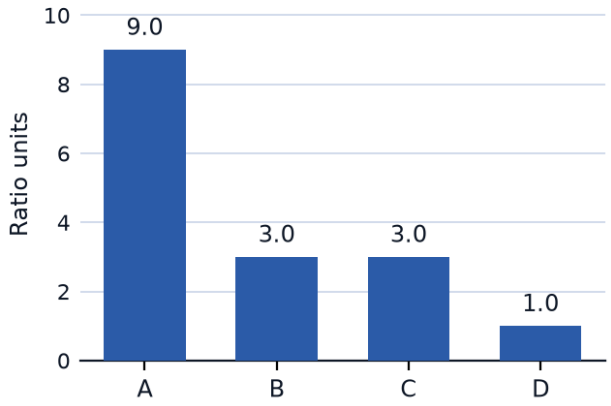
In a two-way table, a row total is 80, a column total is 120, and the grand total is 300. Find the expected count for that cell, using $E = \frac{\text{(row total)(column total)}}{\text{grand total}}$.

13 [Intermediate] • Target: Computing expected counts for a chi-square test

State the degrees of freedom for a chi-square goodness-of-fit test with 6 categories, using $df = (\text{number of categories}) - 1$.

14 [Intermediate] • Target: Computing expected counts for a chi-square test

A genetics experiment expects offspring to appear in a 9:3:3:1 ratio across four phenotype categories, based on 160 total offspring observed. Find the expected count for each category, and briefly describe what the chi-square goodness-of-fit test would examine using these expected counts.



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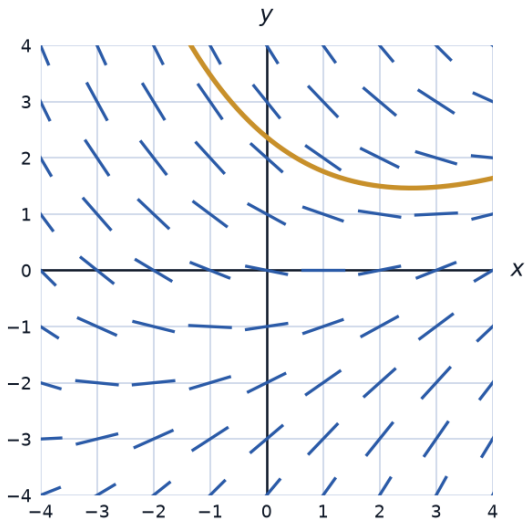
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A study examines the relationship between hours of sleep per night (x) and reaction time in milliseconds (y) for 30 participants, finding a regression slope $b_1 = -8.5$ with standard error $SE_{b_1} = 2.3$. State the hypotheses for testing whether more sleep is associated with a significant decrease in reaction time (a one-sided test), compute the test statistic, and interpret the sign of the slope in context.

16 **[Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A university surveys 240 students, recording both their preferred study location (library, dorm, coffee shop) and their class year (underclassman, upperclassman), to test whether study location preference is independent of class year. The resulting chi-square test statistic is $\chi^2 = 11.4$ with $df = 2$, and the corresponding p-value is 0.0033. State the conclusion at $\alpha = 0.01$, and explain, in the context of this scenario, what rejecting or failing to reject H_0 would mean for the university's understanding of its students.



17 **[Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A researcher studies the relationship between advertising spending (in thousands of dollars, x) and monthly sales (in thousands of dollars, y) for 22 small businesses, obtaining a regression slope $b_1 = 3.2$ with standard error $SE_{b_1} = 0.85$. Construct a 95% confidence interval for the true population slope β_1 using a critical value $t^* \approx 2.086$ (with $df = 20$), and interpret the interval in context, including whether it suggests advertising spending has a statistically significant positive association with sales.

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18 [Challenging] • Target: Chi-square test for independence • Cross-Lesson: 3.03 The Normal Distribution

Level 2 · MediumA significance test produces a test statistic of $z = 2.15$ for a two-sided test $H_0 : p = 0.40$ vs $H_a : peq 0.40$. What is the p-value, and what decision is made at $\alpha = 0.05$?

19 [Challenging] • Target: Interpreting chi-square results • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Eight patients with chronic insomnia try a new relaxation therapy. Their sleep duration (hours/night) is measured before and after 4 weeks of therapy: Patient differences $d = \text{After} - \text{Before}$ yield $\bar{x}_d = 1.25$ hours and $s_d = 0.80$ hours. Assuming normality of differences, test whether the therapy significantly increased sleep at $\alpha = 0.01$.

20 [Challenging] • Target: Inference for the slope of a regression line • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

[Unit 4 · Statistical Inference: Proportions, Means & Slopes · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Chi-square goodness-of-fit test

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $17, 21, 13, 9$. Compute the chi-square test statistic using $\chi^2 = \sum (O - E)^2 / E$.

a. 5.33

b. 4.33

c. 5.00

d. 6.33

22 [Foundational] • Target: Chi-square test for independence

In a two-way table, a row total is 87, a column total is 46, and the grand total is 200. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 23.01

b. 112.99

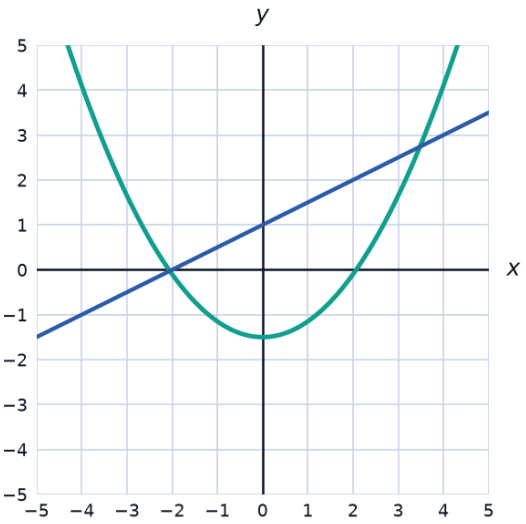
c. 20.01

d. 17.01

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23 [Foundational] • Target: Interpreting chi-square results

A regression gives a sample slope $b_1 = -3.5$ with standard error $SE_{b_1} = 1.1$, based on data with $df = 10$. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.228$.



a. $(-5.45, -1.05)$

b. $(-5.95, -1.05)$

c. $(-6.95, -0.05)$

d. $(-5.95, -1.55)$

24 [Foundational] • Target: Inference for the slope of a regression line

A regression gives a sample slope $b_1 = 3.8$ with standard error $SE_{b_1} = 0.9$. Compute the test statistic for testing $H_0 : \beta_1 = 0$, using $t = b_1/SE_{b_1}$, to two decimal places.

a. 3.22

b. 4.22

c. 5.22

d. −4.22

25 [Foundational] • Target: Conditions for regression inference

A chi-square test produces a test statistic $\chi^2 = 15.9$ with $df = 4$, and a critical value of $\chi^2_{0.05} = 9.49$ at $\alpha = 0.05$. State the conclusion, comparing the test statistic directly to the critical value.

a. The test cannot be completed without a p-value

b. Reject H_0; there is convincing evidence of an association

c. Fail to reject H_0; there is not convincing evidence of an association

d. Accept H_0 as definitely true

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26 [Foundational] • Target: Putting it together: a full inference scenario

A chi-square goodness-of-fit test has 4 categories with observed counts 21, 25, 50, 10 (equal expected proportions). Find the chi-square test statistic.

a. 32.34

b. 35.34

c. 30.34

d. 36.34

27 [Foundational] • Target: Chi-square test for homogeneity

A study investigates the effectiveness of a new teaching method by comparing the test scores of students who used the new method versus those who used the traditional method. The results are summarized in a two-way table. If the table has 4 rows and 3 columns, what is the degrees of freedom (df) for a chi-square test of homogeneity?

a. 12

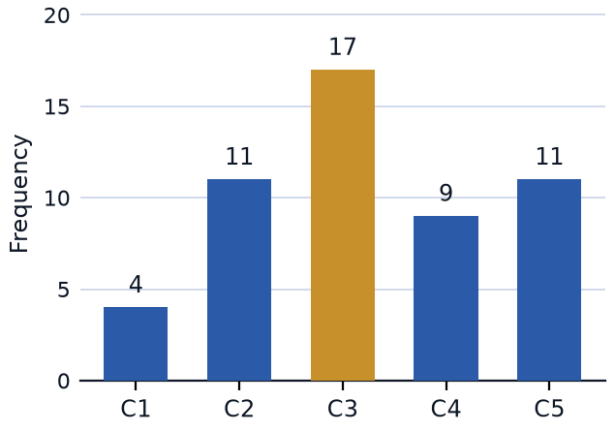
b. 6

c. 5

d. 2

28 [Foundational] • Target: Computing expected counts for a chi-square test

A chi-square goodness-of-fit test has 4 categories. Find the degrees of freedom.



a. 4

b. 2

c. 5

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Chi-square goodness-of-fit test

A chi-square test of independence uses a 2×4 table. Find the degrees of freedom.

a. 4

b. 5

c. 2

d. 3

30 [Intermediate] • Target: Chi-square test for independence

In a chi-square test of independence for a contingency table with 7 rows and 2 columns (Trial #5), what is the value of df?

a. 6

b. 7

c. 14

d. 13

31 [Intermediate] • Target: Interpreting chi-square results

In a test of independence with total sample $N = 390$, row total 85, and column total 70 (Experiment 5), find the expected cell count.

a. 14.38

b. 16.33

c. 15.26

d. 44.29

32 [Intermediate] • Target: Inference for the slope of a regression line

Chi-square test is used for:

a. Comparing two means

b. Relationships between categorical variables

c. Regression coefficients

d. Differences in variances only

33 [Intermediate] • Target: Conditions for regression inference

Conditions for inference with proportions (1-prop z-test):

a. n>30

b. np≥10 and n(1-p)≥10

c. Normal population

d. Known σ

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34 [Intermediate] • Target: Putting it together: a full inference scenario

A test of $H_0 : \mu = 81$ uses a sample of $n = 10$ with $\bar{x} = 85$, $s = 12$. Find the test statistic t.

a. 2.05

b. 3.05

c. 1.05

d. 1.55

35 [Intermediate] • Target: Chi-square test for homogeneity

In a hypothesis test, failing to reject H_0 when H_a is actually true. What is this called?

a. Confounding

b. Type II error

c. Type I error

d. Sampling bias

36 [Intermediate] • Target: Computing expected counts for a chi-square test

In a hypothesis test, rejecting H_0 when H_0 is actually true. What is this called?

a. Confounding

b. Type I error

c. Type II error

d. Sampling bias

37 [Intermediate] • Target: Chi-square goodness-of-fit test

A pharmaceutical company is investigating the efficacy of a new drug to reduce blood pressure. They want to design a randomized controlled experiment. Participants will be recruited from a specific patient population. To ensure generalizability and account for potential confounding factors, the researchers decide to stratify the participants by age group (Under 40, 40-60, Over 60) and by the severity of their hypertension (Mild, Moderate, Severe). If there are 150 participants in total and an equal number of participants are to be assigned to each treatment group (Drug A, Drug B, Placebo), and the number of participants in each stratum is proportional to the expected prevalence in the general population, how many participants would be allocated to the stratum 'Under 40' and 'Mild' hypertension if the proportion of individuals in this stratum is 0.08?

a. 10

b. 12

c. 15

d. 8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Chi-square test for independence • Cross-Lesson: 3.03 The Normal Distribution

What is the general structure of a 90% confidence interval?

a. estimate × (critical value) + (standard error)

b. (critical value) ± (standard error) × estimate

c. estimate – (critical value) ÷ (standard error)

d. estimate ± (critical value) × (standard error)

39 [Challenging] • Target: Interpreting chi-square results • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Increasing significance level α:

a. Decreases Type I error

b. Increases power

c. Decreases power

d. Has no effect

40 [Challenging] • Target: Inference for the slope of a regression line • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A 95% confidence interval means:

a. 95% chance the true parameter is inside

b. 95% of all possible such intervals contain the true value

c. The sample mean is 95% accurate

d. p-value is 0.05

41 [Challenging] • Target: Conditions for regression inference • Cross-Lesson: 1.01 Measures of Center & Spread

p-value is the probability of:

a. The null hypothesis being true

b. Observing data at least as extreme, assuming H■ true

c. The alternative hypothesis being true

d. A Type II error

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42 [Challenging] • Target: Putting it together: a full inference scenario • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The scatterplot below shows the number of hours a group of students studied for a test and their scores on that test. The line of best fit for the data is given by the equation $S = 5H + 40$, where S is the score on the test and H is the number of hours studied. If a student studied for x hours and achieved a score of 85, what is the value of x?

a. 10

b. 9

c. 8

d. 7

43 [Challenging] • Target: Chi-square test for homogeneity • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

The scatterplot below shows the number of hours spent studying for a test versus the score achieved on that test for a group of students. The equation of the line of best fit is given by $S = 5H + 50$, where S is the score and H is the number of hours spent studying. If a student scored 80 on the test, approximately how many hours did they study?

a. 5.0 hours

b. 5.5 hours

c. 6.0 hours

d. 6.5 hours

44 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 2.01 Sampling Methods & Bias

A line of best fit relating hours studied (x) to test score ($\hat{y}$) is $\hat{y} = 10x + 27$. Using this model, what is the predicted score for a student who studies 4 hours?

a. 63

b. 67

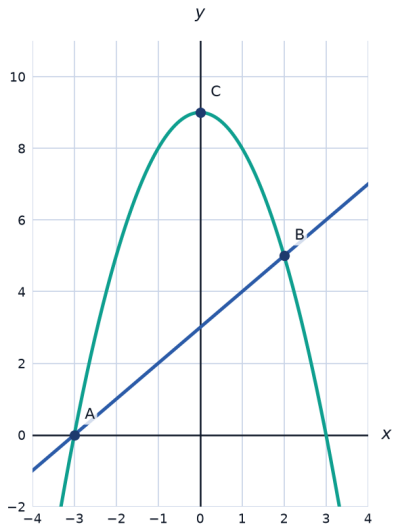
c. 62

d. 76

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45 [Challenging] • Target: Chi-square goodness-of-fit test • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of ΔABC?



a. 12.5

b. 15

c. 22.5

d. 30

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